

Practical Workbook
CT-361
Artificial Intelligence & Expert Systems



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Q1. Write a Python program that asks the user to enter their name, age, department, and semester, and displays all the information.

SOLUTION:

```
1 print(input("Enter your name:"))
2 print(input("Enter your age:"))
3 print(input("Enter your department:"))
4 print(input("Enter your semester:"))
```

OUTPUT:

```
PS C:\Users\USER\Desktop\AIES\AIES Labs\LAB#06> python Lab6q1.py
Enter your name:Moazzam Farooqui
Moazzam Farooqui
Enter your age:19
19
Enter your department:CSIT
CSIT
Enter your semester:5th
5th
```

Q2. Basic Calculator

Write a Python program that takes two numbers from the user and displays their:

- Addition
- Subtraction
- Multiplication
- Division

SOLUTION:

```
1 num1=float(input("Enter number 1:"))
2 num2=float(input("Enter number 2:"))
3
4 print("Result is:", num1+num2)
5 print("Result is:", num1-num2)
6 print("Result is:", num1*num2)
7 print("Result is:", num1/num2)
```

OUTPUT:

```
PS C:\Users\USER\Desktop\AIIES\AIIES Labs\LAB#06> python Lab6q2.py
Enter number 1:5
Enter number 2:4
Result is: 9.0
Result is: 1.0
Result is: 20.0
Result is: 1.25
```

Q3. Area of a Rectangle

Write a Python program that takes the length and width of a rectangle and calculates its area and perimeter.

SOLUTION:

```
1 length=float(input("Enter length of rectangle:"))
2 width=float(input("Enter width of rectangle:"))
3 area=length*width
4 perimeter=2*(length+width)
5 print("Area of Rectangle:",area)
6 print("Perimeter of Rectangle:",perimeter)
```

OUTPUT:

```
PS C:\Users\USER\Desktop\AIIES\AIIES Labs\LAB#06> python Lab6q3.py
Enter length of rectangle:6.5
Enter width of rectangle:4.5
Area of Rectangle: 29.25
Perimeter of Rectangle: 22.0
```

Q4. Temperature Converter

Write a Python program that takes temperature in Celsius and converts it into Fahrenheit.

SOLUTION:

```
1 temperature=float(input("Enter temperature in Celsius:"))
2 fahrenheit=(temperature*9/5)+32
3 print("Temperature in Fahrenheit:",fahrenheit)
```

OUTPUT:

```
PS C:\Users\USER\Desktop\AIIES\AIIES Labs\LAB#06> python Lab6q4.py
Enter temperature in Celsius:31.2
Temperature in Fahrenheit: 88.16
```

Q5. Average Marks

Write a Python program that takes marks of five subjects from the user and calculates the total marks and average marks.

SOLUTION:

```
1  subject1=float(input("Enter subject 1 marks:"))
2  subject2=float(input("Enter subject 2 marks:"))
3  subject3=float(input("Enter subject 3 marks:"))
4  subject4=float(input("Enter subject 4 marks:"))
5  subject5=float(input("Enter subject 5 marks:"))
6
7  total=subject1+subject2+subject3+subject4+subject5
8  average=total/5.0
9
10 print("Total marks=",total)
11 print("Average marks=",average)
```

OUTPUT:

```
PS C:\Users\USER\Desktop\AIIES\AIIES Labs\LAB#06> python Lab6q5.py
Enter subject 1 marks:45.5
Enter subject 2 marks:46.0
Enter subject 3 marks:75.5
Enter subject 4 marks:67.0
Enter subject 5 marks:55.0
Total marks= 289.0
Average marks= 57.8
```

Q6. Shopping Bill

Write a Python program that takes the name, quantity, and price of three products and calculates the total bill.

SOLUTION:

```
1  product1_name=input("Enter product 1 name:")
2  product1_qty=int(input("Enter product 1 quantity:"))
3  product1_price=int(input("Enter price of product 1:"))
4
5  product2_name=input("Enter product 2 name:")
6  product2_qty=int(input("Enter product 2 quantity:"))
7  product2_price=int(input("Enter price of product 2:"))
8
9  product3_name=input("Enter product 3 name:")
10 product3_qty=int(input("Enter product 3 quantity:"))
11 product3_price=int(input("Enter price of product 3:"))
12
13 totalbill=(product1_qty*product1_price)+(product2_qty*product2_price)+(product3_qty*product3_price)
14 print("Total bill=",totalbill)
```

OUTPUT:

```
PS C:\Users\USER\Desktop\AIIES\AIIES Labs\LAB#06> python Lab6q6.py
Enter product 1 name:Cold drink
Enter product 1 quantity:10
Enter price of product 1:90
Enter product 2 name:Chocolate
Enter product 2 quantity:5
Enter price of product 2:100
Enter product 3 name:Juice
Enter product 3 quantity:6
Enter price of product 3:80
Total bill= 1880
```

Q7. Simple Interest

Write a Python program that takes principal amount, rate, and time and calculates simple interest using:

$$\text{Simple Interest} = (\text{Principal} \times \text{Rate} \times \text{Time}) / 100$$

SOLUTION:

```
1 principal_amount=int(input("Enter principal amount:"))
2 rate=float(input("Enter rate(percentage):"))
3 time=float(input("Enter time(in years):"))
4
5 interest=(principal_amount*rate*time)/100
6 print("Interest amount is:",interest)
```

OUTPUT:

```
PS C:\Users\USER\Desktop\AIES\AIES Labs\LAB#06> python Lab6q7.py
Enter principal amount:100000
Enter rate(percentage):5.0
Enter time(in years):2.0
Interest amount is: 10000.0
```

Q8. Swap Two Variables

Write a Python program to take two numbers from the user and swap their values without using a third variable.

SOLUTION:

```
1 number1=int(input("Enter number 1:"))
2 number2=int(input("Enter number 2:"))
3
4 ✓ def swap(number1,number2):
5     |     return number2,number1
6
7 number1,number2=swap(number1,number2)
8
9 print("After swapping:")
10 print("Number 1:",number1)
11 print("Number 2:",number2)
```

OUTPUT:

```
PS C:\Users\USER\Desktop\AIES\AIES Labs\LAB#06> python Lab6q8.py
Enter number 1:7
Enter number 2:23
After swapping:
Number 1: 23
Number 2: 7
```

Q9. String Information

Write a Python program that takes a person's name as input and displays:

- Name
- Number of characters
- First character
- Last character

SOLUTION:

```
1 name=input("Enter your name:")
2
3 print("Name:",name)
4 print("Number of characters:",len(name))
5 print("First character:",name[0])
6 print("Last character:",name[-1])
```

OUTPUT:

```
PS C:\Users\USER\Desktop\AIES\AIES Labs\LAB#06> python Lab6q9.py
Enter your name:Moazzam
Name: Moazzam
Number of characters: 7
First character: M
Last character: m
```

Q10. Seconds Converter

Write a Python program that takes number of seconds and converts it into minutes and remaining seconds.

SOLUTION:

```
1 seconds=int(input("Enter number of seconds:"))
2
3 minutes=int(seconds/60)
4 remaining_seconds=seconds-(minutes*60)
5
6 print("Minutes:",minutes)
7 print("Remaining seconds:",remaining_seconds)
```

OUTPUT:

```
PS C:\Users\USER\Desktop\AIIES\AIIES Labs\LAB#06> python Lab6q10.py
Enter number of seconds:150
Minutes: 2
Remaining seconds: 30
```

Q11. Employee Salary

Write a Python program that takes an employee's basic salary, house allowance, and medical allowance and calculates the total salary.

SOLUTION:

```
1 basic_salary=float(input("Enter basic salary:"))
2 house_allowance=float(input("Enter house allowance:"))
3 medical_allowance=float(input("Enter medical allowance:"))
4
5 total_salary=basic_salary+house_allowance+medical_allowance
6
7 print("Total salary:",total_salary)
```

OUTPUT:

```
PS C:\Users\USER\Desktop\AIIES\AIIES Labs\LAB#06> python Lab6q11.py
Enter basic salary:100000.0
Enter house allowance:25000.0
Enter medical allowance:10000.0
Total salary: 135000.0
```

Q12. Distance Converter

Write a Python program that takes distance in kilometers and converts it into meters and centimeters.

SOLUTION:

```
1 kilometers=float(input("Enter distance in kilometers:"))
2
3 meters=kilometers*1000
4 centimeters=kilometers*100000
5
6 print("Distance in meters:",meters)
7 print("Distance in centimeters:",centimeters)
```

OUTPUT:

```
PS C:\Users\USER\Desktop\AIIES\AIIES Labs\LAB#06> python Lab6q12.py
Enter distance in kilometers:10.0
Distance in meters: 10000.0
Distance in centimeters: 1000000.0
```

Q13. Restaurant Bill

Write a Python program that takes the prices of three food items, calculates the subtotal, adds a 5% service charge, and displays the final bill.

SOLUTION:

```
1 item1=float(input("Enter price of food item 1:"))
2 item2=float(input("Enter price of food item 2:"))
3 item3=float(input("Enter price of food item 3:"))
4
5 subtotal=item1+item2+item3
6 service_charge=0.05*subtotal
7 total=subtotal+service_charge
8
9 print("Final bill is:",total)
```

OUTPUT:

```
PS C:\Users\USER\Desktop\AIES\AIES Labs\LAB#06> python Lab6q13.py
Enter price of food item 1:750.0
Enter price of food item 2:1000.0
Enter price of food item 3:1500.0
Final bill is: 3412.5
```